Worksheet 2-3 A
Estimating Derivatives of a Function at a Point #1
AP Calculus Name_____

For 1 – 3, explain the meaning of the following statements.

1.) $f'(-2) = 4$ 2.) $g'(0) = -5$ 3.) $h'(3) = 0$

Use the following tables to approximate the given derivative values.

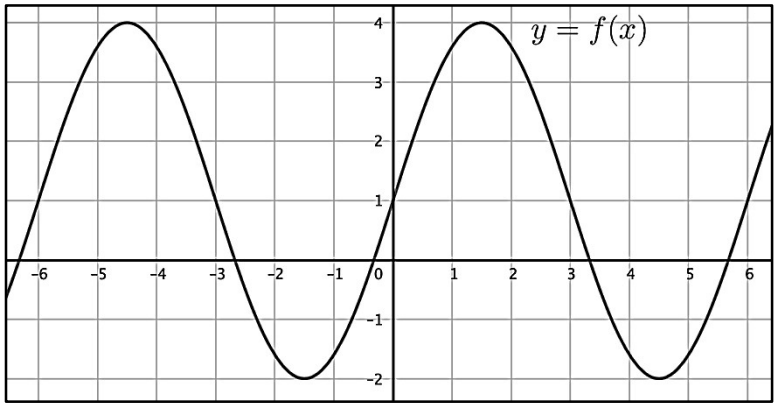
x	0	2	6	7
$h(x)$	1	-5	3	2

4.) $h'(1)$ 5.) $h'(4)$

x	-3	-1	3	5
$g(x)$	4	2	-1	-1

6. $g'(-2)$ 7. $g'(4)$

Use the following graph to approximate the given values.



8.) $f'(-4)$

9.) $f'(0)$

10.) $f'(4.5)$

11.) $f(1)$

12.) $f'(-3)$

13.) $f(3)$

Use a calculator to approximate the following values.

$$f(x) = 2.3\sin(x^2 - 3x + 7) - 5$$

14.) $f'(2)$ 15.) $f'\left(\frac{\pi}{7}\right)$ 16.) $f(-5)$ 17.) $f'(0)$

$$g(x) = \frac{3 - x^2}{\ln|4x - 7|}$$

18.) $g'(3)$ 19.) $g'(-6)$ 20.) $g(0)$ 21.) $g'(e^2)$